Calculus II

Tutorial Problems #2 24–28 Feb 2025

◇ **Topics:**

- Limits (continued)
- Limit along paths
- Continuity

In Problems 1–2, by considering different paths of approach, show that the functions have no limit as $(x, y) \to (0, 0)$.

Problem 1.

$$f(x, y) = -\frac{x}{\sqrt{x^2 + y^2}}.$$

Problem 2.

$$f(x, y) = \frac{xy}{|xy|}.$$

Continuous Extension. In Problems 3–5, define $f(0, 0)$ in a way that extends f to be continuous at the origin.

Problem 3.

$$f(x, y) = xy\frac{x^2 - y^2}{x^2 + y^2}.$$

Problem 4.

$$f(x, y) = \ln\left(\frac{3x^2 - x^2y^2 + 3y^2}{x^2 + y^2}\right).$$

Problem 5.

$$f(x, y) = \frac{3x^2y}{x^2 + y^2}.$$

In Problems 6–7, using the limit definition show the following assertions

Problem 6. The function

$$f(x, y, z) = x + y - z$$

is continuous at every point (x_0, y_0, z_0).

Problem 7. The function

$$f(x, y, z) = x^2 + y^2 + z^2$$

is continuous at the origin.

Problem 8. Investigate at what points (x, y) in the plane the functions are continuous.

(a)

$$f(x, y) = \sin(x + y)$$

(b)

$$f(x, y) = \ln(x^2 + y^2)$$

Problem 9. Investigate at what points (x, y, z) in space the functions are continuous.

(a)

$$h(x, y, z) = xy \sin \frac{1}{z}$$

(b)

$$h(x, y, z) = \frac{1}{x^2 + z^2 - 1}$$

Problem 10. Let

$$f(x, y) = \begin{cases} 1, & y \geq x^4 \\ 1, & y \leq 0 \\ 0, & \text{otherwise.} \end{cases}$$

Find each of the following limits, or explain why the limit does not exist.

(a)

$$\lim_{(x,y)\to(0,1)} f(x, y).$$

(b)

$$\lim_{(x,y)\to(2,3)} f(x, y).$$

(c)

$$\lim_{(x,y)\to(0,0)} f(x, y).$$

END

Answers to selected problems

3. $f(0,0) = 0$.

4. $\ln 3$

5. $f(0,0) = 0$

8. (a) $\mathbb{R}$ (b) $\{(x,y) \in \mathbb{R}^2;\ (x,y) \neq (0,0)\}$

9. (a) $\{(x,y,z) \in \mathbb{R}^3 :\ z \neq 0\}$ (b) $\{(x,y,z) \in \mathbb{R}^3 :\ x^2 + z^2 - 1 \neq 0\}$

10. (a) 1 (b) 0 (c) Does not exist.